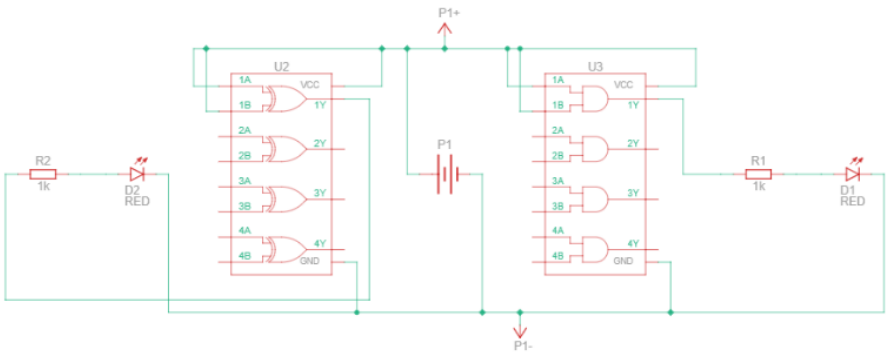
CSO Lab 2: Combinational Circuit Design

Observation:

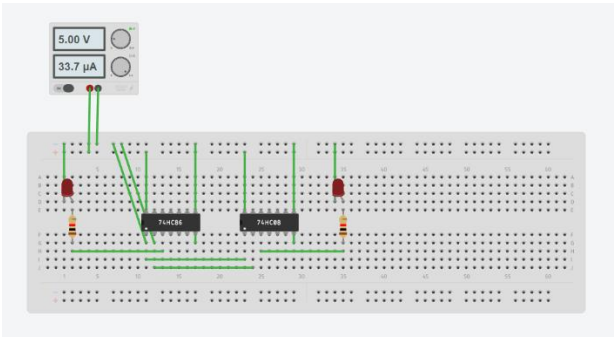
Half Adder

INPUTS		OUTPUTS	
A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

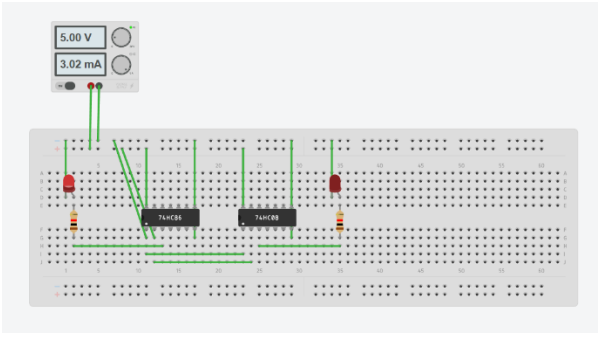
Schematic diagram



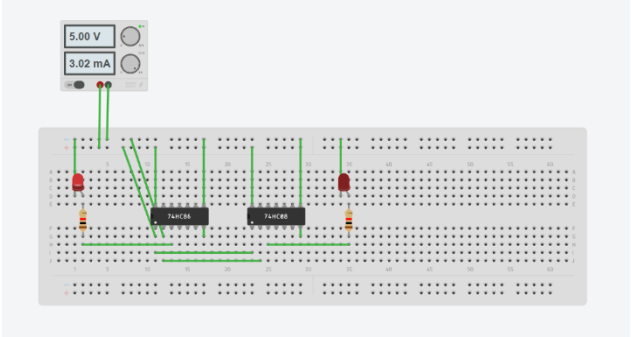
Input: A=0, B=0 Output: S=0, C=0



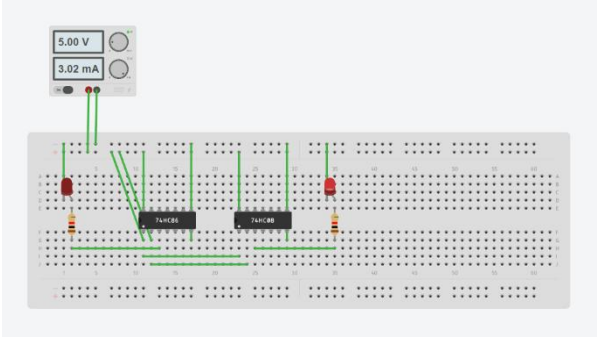
Input: A=0, B=1 Output: S=1,C=0



Input: A=1, B=0 Output: S=1,C=0



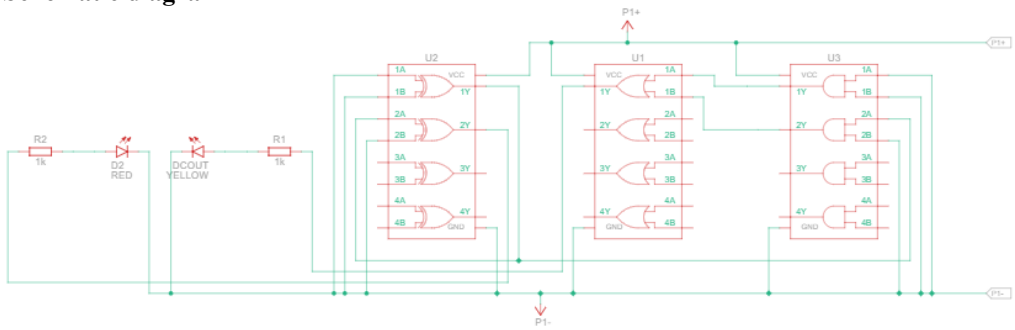
Input: A=1, B=1 Output: 1



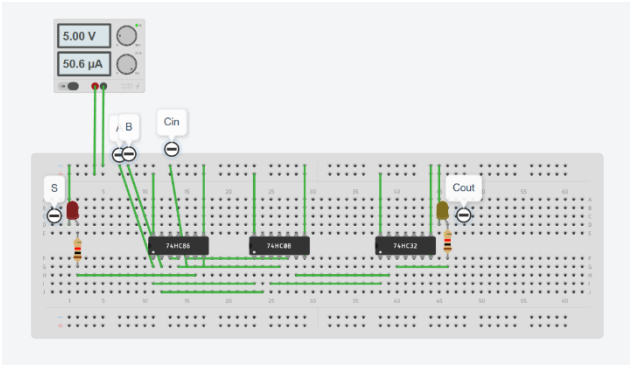
Full Adder

INPUTS			OUTPUTS	
A	B	CARRY IN	SUM	CARRY OUT
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

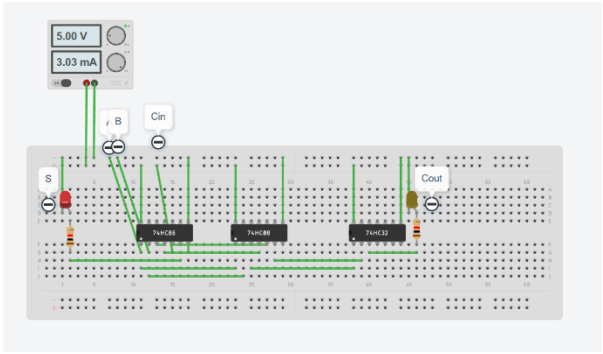
Schematic diagram



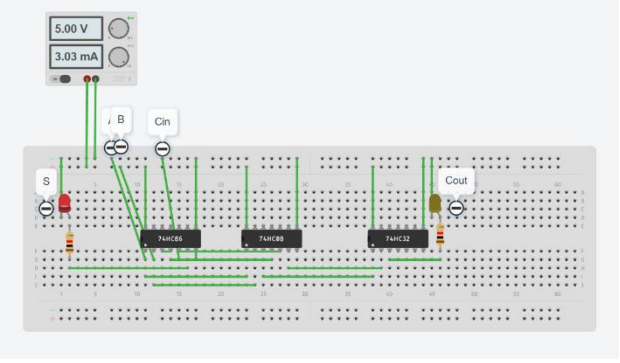
Input: A=0 B=0 Cin=0 Output: S=0, Cout=0



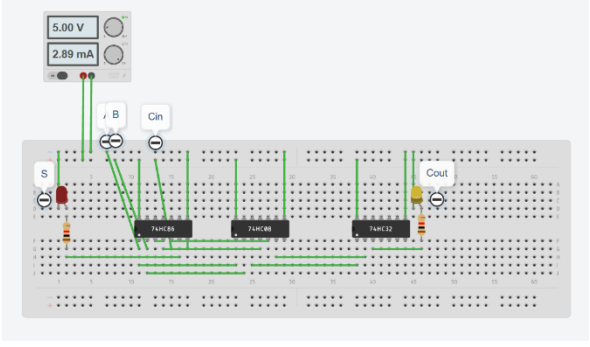
Input: A=0 B=0 Cin=1 Output: S=1, Cout=0



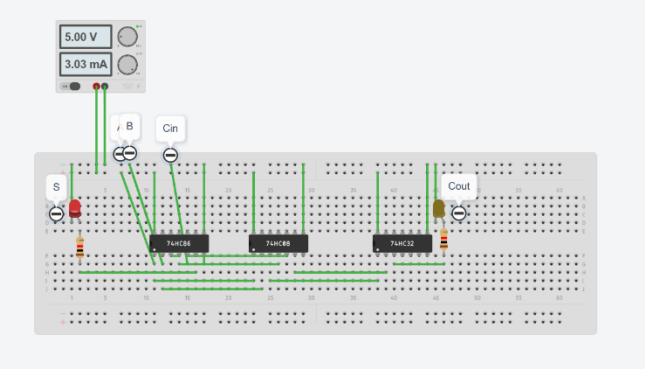
Input: A=0 B=1 Cin=0 Output: S=1, Cout=0



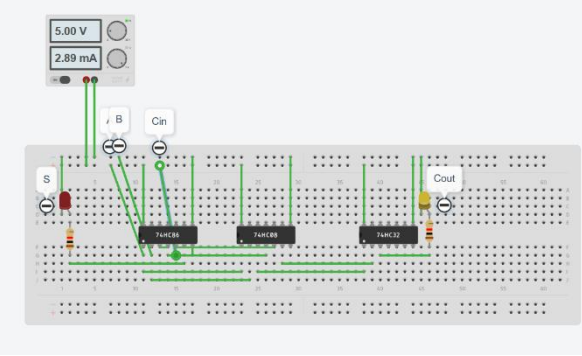
Input: A=0 B=1 Cin=1 Output: S=0, Cout=1



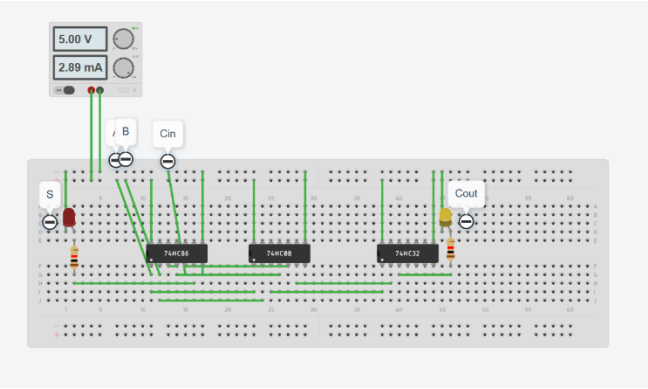
Input: A=1 B=0 Cin=0 Output: S=1, Cout=0



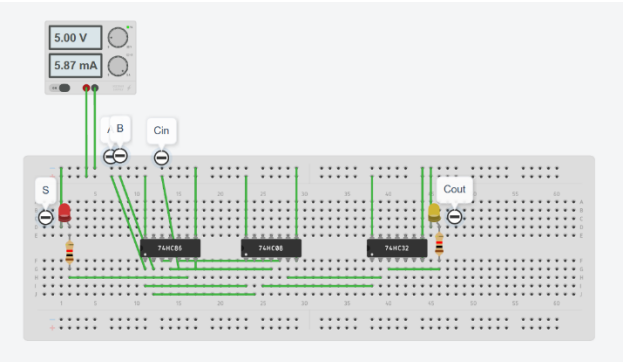
Input: A=1 B=0 Cin=1 Output: S=0, Cout=1



Input: A=1 B=1 Cin=0 Output: S=0, Cout=1



Input: A=1 B=1 Cin=1 Output: S=1, Cout=1



Result:

1.Half adder is made up of one XOR gate and one AND gate
2.Full adder is made up of two half adder and one OR gate.